

Multi-Agent Systems (MAS) for Sustainable Smart Farming: A Systematic Literature Review (SLR) of Cooperative Irrigation, Resource Allocation, Livestock Management, and Environmental Sustainability

Chayan Talukder [0001172880] — chayan.talukder@studio.unibo.it

Abstract

Modern agriculture requires simultaneous decisions concerning irrigation, crop management, livestock welfare, weather variability, energy use, and the allocation of limited resources. These characteristics are strongly interconnected, making it a suitable domain for MAS, in which autonomous and heterogeneous agents perceive local conditions, pursue individual or shared objectives, communicate, cooperate, and negotiate over resources.

The sustainability dimension is important as agriculture accounts for approximately 72% of global freshwater withdrawals, making water allocation and irrigation major environmental concerns. The IPCC estimates 13–21% of global anthropogenic GHG emissions, nutrient runoff, water pollution, soil degradation, and ecosystem pressure.

Existing research has investigated MAS for these areas; however, they are often studied separately rather than an integrated farm-level decision problem. This project therefore proposes a SLR investigating how MAS have been applied to integrated smart-farm decision-making.

Aim and Research Contribution

Primary Research Question: How are Multi-Agent Systems designed and used to coordinate agricultural decision making and shared-resource allocation across crop, irrigation, and livestock systems, and to what extent do existing approaches address environmental sustainability?

The review will analyze the literature through the following connected dimensions:

Agent architecture → Communication and Coordination → Resource Management → Agricultural Outcomes → Environmental outcomes

Conceptual Scope of the Review

The general MAS decision cycle will also be used as a classification framework:

Environment / Sensors → Agent Perception → Local State / Beliefs → Need, Goal or Utility Assessment → Agent Communication → Coordination / Negotiation → Resource Allocation / Action → Farm + Environmental Outcomes → Updated Environment

Systematic Review Methodology

- **Information Sources**
- **Indicative Master Search Strategy**
- **Database-Specific Syntax Adaptation**
- **Publication Window:** 2010-2026 (Approx)
- **Screening Threshold:** maximize reasonable coverage

Reference Management and PRISMA Workflow

Two complementary tools will be used rather than multiple overlapping reference managers:

- **Rayyan** — SLR Screening and PRISMA Workflow
- **Zotero** — Reference Library

Quality Assessment and Data Extraction

Each included study will be evaluated for methodological quality using criteria such as: research objectives, roles and autonomy, communication and coordination mechanisms, environment and data, reproducibility, limitations

Analysis and Synthesis

The literature will be compared across coordination strategies analogous to the original smart-farm concept:

- Independent/decentralized decision-making
- Centralized or hierarchical coordination
- Cooperative/negotiated multi-agent allocation
- Learning or adaptive approaches, where present

Reported outcomes will be mapped to common metric groups. The review will also examine scenarios corresponding to the proposed smart-farm system.

Single-Reviewer Consideration

As the project will be conducted by one researcher, study selection will use a **single-reviewer screening** process. This will be acknowledged as a methodological limitation.

Expected Outcomes

- a reproducible systematic review of MAS in smart agriculture
- a taxonomy of agricultural agents, architectures, and coordination mechanisms
- a detailed analysis of irrigation and shared-resource negotiation
- a synthesis of MAS and AI approaches to livestock management
- an environmental-sustainability evidence map
- comparison of independent, centralized, cooperative, and learning-based approaches
- identification of common datasets, scenarios, evaluation metrics, and deployment practices
- identification of technical and scientific research gaps

Proposed Timeline

SEPTEMBER 2026

- Finalize protocol and research questions
- Pilot and refine search strings
- Database searches

OCTOBER 2026

- Data extraction, taxonomy and evidence synthesis
- Environmental sustainability analysis
- Research-gap identification
- PRISMA documentation
- Final report and presentation